Journal of Computational Social Science (2026) 9:17
https://doi.org/10.1007/s42001-025-00452-1

RESEARCH ARTICLE

ChatGPT as a research proxy: simulating human attitudes in social science research

Antonina Rafikova[1] · **Anatoly Voronin**[1]

Received: 22 July 2025 / Accepted: 1 December 2025
© The Author(s), under exclusive licence to Springer Nature Singapore Pte Ltd. 2025

Abstract
As artificial intelligence transforms the landscape of social science research, large language models (LLMs) like ChatGPT present both unprecedented opportunities and unprecedented challenges. This study explores the application of ChatGPT as "surrogates" or computational substitutes for human participants in sociological and socio-psychological research. By simulating responses to complex socio-cultural issues, we investigate how well ChatGPT can replicate human attitudes toward immigration, gender stereotypes, and LGB parenting attitudes. We utilized a general simulation model employing detailed demographic prompts to generate synthetic participant responses, assessing their accuracy and political biases. When addressing topics in selected socio-psychological domains (gender, migration, and family values), ChatGPT tended to produce responses aligned with liberal-prosocial norms, which likely reflect alignment and safety-training objectives rather than stable ideological commitments. Across all simulations, ChatGPT reproduced the direction and magnitude of human attitudinal trends with high predictive accuracy (low MAE and MAPE). However, explanatory power (R^2) remained modest across models, indicating that the LLM can reliably approximate aggregate patterns but not the deeper structure of attitudinal variance. These findings suggest that LLMs may serve as useful proxy participants for distributional replication, though not yet for mechanistic inference in attitudinal research. Since GPT-based models are trained predominantly on Western and Anglophone textual data, their simulated attitudes may reflect culturally specific semantic patterns rather than globally representative ones. Thus, the present findings illustrate the potential of LLMs for attitudinal simulation within Western-linguistic contexts, while underscoring the need for culturally diversified models in future research. This research underscores the necessity for critical evaluation of AI-generated data in social science contexts and calls for further refinement of LLM methodologies.

Keywords ChatGPT · Artificial intelligence · Computational social science · Large language models · AI simulation

Extended author information available on the last page of the article

Published online: 07 January 2026

1 Introduction

As artificial intelligence (AI) continues to evolve, it is reshaping various fields, including social science research. ChatGPT manifests as computational semi-human [2] and is able to combine features such as emotional expression, perceptual skills, and a nuanced understanding of context [26]. Recent progress in large language models (LLMs) pre-trained on extensive datasets of text has initiated a groundbreaking phase in the social science research [4]. AI tools can be potential solutions to a range of research challenges, such as time constraints, budget limitations, and cognitive constraints, which could allow scientists to increase productivity and objectivity [33]. Four distinct visions of AI for scientific research have been identified by Messeri and Crockett [33]: Oracle, Surrogate, Quant, and Arbiter. Our study investigates the vision of AI as Surrogate for scientific research.

According to Dillion et al. [13], LLMs can capture the "wisdom of the crowd", which posits that the collective average of numerous predictions tends to outperform individual guesses taken independently [50]. The behavioral patterns exhibited by LLMs show a noteworthy alignment with the collective behavior of human participants in published psychology research [52, 54]. This phenomenon may be attributed to the training processes utilized for LLMs: "The 'minds' of language models are trained on vast amounts of human expression, so their expressions can indirectly capture millions of human minds" ([4], p. 3). Consequently, when psychologists gather assessments or "norms" for their stimuli, they generally obtain multiple ratings for each linguistic stimulus and subsequently compute the mean of these ratings. This approach serves as a fundamental principle underlying ensemble models in machine learning [12, 31, 53].

Generative AI models seem able to mimic human behavior in certain contexts, and this might prove valuable to social scientists in the context of their research purposes [4, 20]. The concept of employing LLMs as in silica "model organisms" [53] has attracted significant attention [1, 3, 7, 13, 18, 21, 24, 27, 47]. Argyle et al. [3] conceptualized algorithmic fidelity and introduced silicon sampling, "a methodology by which a language model can generate a virtual population of respondents, correcting skewed marginal statistics in the training data" ([3], p. 338). Algorithmic fidelity is a "fine-grained and demographically correlated property of LLM, meaning that proper conditioning will cause it to accurately emulate response distributions from a wide variety of human subgroups" ([3], p. 337). The importance of high algorithmic fidelity in language models cannot be overstated for their application in the social sciences, as it allows researchers to derive insights from a single model regarding the diverse patterns of attitudes and perspectives prevalent among various groups (such as women, men, White individuals, etc.) as well as the intersections and combinations of these groups (female Republicans, White males, etc.). It is important to recognize that while language models are unlikely to completely replace human participants, they may serve as valuable tools throughout different stages of the research process, like in idea generation and refinement, item piloting, and as an additional check for robustness and reliability [13]. Dillion et al. [13] encourage scientists to compare simulated language model data with human data to see how aligned they are across different domains and populations. While recent work has demonstrated that LLM

can reproduce population-level patterns with impressive precision, their explanatory validity remains limited. In our study, the models captured the mean direction and variability of attitudes with high predictive accuracy, yet the explanatory power (R^2) was consistently modest. This distinction underscores that LLMs currently function as statistical proxies of human populations rather than as cognitive analogs capable of modeling latent psychological processes. At the same time, the cultural and linguistic asymmetry of LLM training data imposes significant constraints on the generalizability of such simulations. Because GPT-based models are primarily trained on Western and English-language corpora, they tend to internalize Western normative and semantic frameworks. Consequently, the validity of simulated attitudinal distributions should be interpreted within the scope of Western-linguistic contexts rather than as universal patterns.

One notable difference between human participants and LLM-generated synthetic subjects is that each LLM serves as a single participant [13]. Although language models are trained on data from a wide array of individuals, they tend to consolidate the variability of opinions into a single dominant viewpoint [45]. Bisbee et al. [7] demonstrated that the performance of synthetic data is influenced by the attributes present in the prompt, revealing that the majority of the outcomes are significantly affected by political covariates. Including in prompts detailed information about the characteristics of a respondent makes LLMs respond in a manner that is very similar to real respondents with the same attributes [3]. Argyle et al. [3] conclude that "by conditioning the model on thousands of socioeconomic backstories from real human[s]", LLMs are able to generate synthetic opinions that "reflect[] the complex interplay between ideas, attitudes, and sociocultural context that characterize human attitudes". Argyle et al. [3] empirically demonstrated that including demographic attributes in prompts fosters GPT-3's ability to simulate human participants in predicting voting choices. Niszczota and Janczak (2023) proved that GPT-4 could replicate the cross-cultural differences for each personality factor through large-scale experiments. Rao et al. [39] and Serapio-García et al. [47] emphasized the importance of considering context and specific prompting configuration when using ChatGPT to emulate diverse human behaviors.

Xu et al. [55] proposed a framework for categorizing explorations in the combination of AI and social science into two directions. The first direction is focused on AI for social science, where AI is utilized as a powerful tool to enhance various stages of social science research. One of these applications, using LLMs for sampling as proxies for specific human subgroups, is a part of the hypothesis verification stage. The second direction is the social science of AI, which explores and understands AI agents as social entities from the viewpoints of psychology, sociology, economics, politics, and linguistics.

One of the most debated issues that LLM-based chatbots raise is political bias. Political leanings refer to a LLM-based agent's preference for certain political beliefs, values, or views [55]. Studies utilizing political orientation survey questions directed at ChatGPT identified a discernible left, liberal, and libertarian (vs. authoritarian) bias [15, 16, 22, 32, 34, 40–42] (Gover, 2023; Van den Broek, 2023). Apart from political bias, ChatGPT's personality traits can influence the accuracy of simulating sociological and psychological studies. Personality ing of GPT models demonstrates

that these LLMs show a prosocial profile [8], perceive itself as highly open [42] and agreeable [8, 25, 28, 42], and have an ENFJ personality type [42]. As stated by Bodroža et al. [8], the prosocial personality characteristics exhibited by LLMs can be attributed to their foundational purpose, which is designed to serve and assist humans in different areas. In the context of mitigating hate speech and reducing social toxicity in their interactions, this personality profile can be viewed as the most appropriate and effective.

A central component of this alignment process is *Reinforcement Learning from Human Feedback* (RLHF), a post-training optimization procedure that adjusts model behavior according to human evaluators' preferences [57]. In RLHF, annotators compare alternative model responses and select those that appear most helpful, harmless, and honest. A separate reward model is then trained to predict these human preferences, and the base language model is fine-tuned through reinforcement learning to maximize predicted rewards. While this procedure effectively reduces toxicity, discrimination, and misinformation, it also embeds the value orientations of the annotator community—typically reflecting liberal–prosocial norms dominant in Western academic and technological environments [40, 45]. Consequently, apparent political or moral biases in ChatGPT may largely emerge as side effects of safety alignment rather than as manifestations of stable ideological positions.

The overarching objective of alignment and fine-tuning is to ensure that model outputs conform to three principal dimensions of desirable behavior: helpfulness, harmlessness, and honesty [57]. During fine-tuning, responses that exemplify these qualities are rewarded, whereas those perceived as misleading, toxic, or discriminatory receive negative feedback. This process gradually shifts the model's probability distributions toward inclusive and norm-conforming expressions, creating what can be described as alignment-induced preference gradients.

From a socio-cognitive standpoint, these gradients function analogously to social desirability mechanisms in human respondents: they encourage prosocial and low-conflict formulations of opinion-like statements while suppressing overtly polarized or exclusionary wording. As a result, model behavior in value-laden domains (e.g., gender, migration, or family norms) reflects the cumulative effects of alignment optimization rather than an autonomous ideological orientation.

The question domains selected for our study—migration attitudes, gender stereotypes, and same-sex parenting—reflect canonical areas of socio-attitudinal bias that have repeatedly been investigated in both psychology and AI alignment research [22, 40, 45]. These domains are particularly relevant because RLHF and alignment procedures explicitly target the minimization of discriminatory, exclusionary, and politically sensitive outputs in precisely these contexts [58]. Examining ChatGPT's behavior in these "high-alignment" domains allows us to observe not only whether the model reproduces human attitudinal patterns, but also how its safety optimization manifests in social-psychological constructs where bias suppression is expected to be strongest.

In addition to alignment mechanisms, the communicative framing of ChatGPT as a helpful assistant also contributes to its response profile. This personality design functions as a socio-pragmatic frame that constrains how the model expresses judgments and evaluations. By emphasizing cooperation, empathy, and politeness, the

"assistant persona" encourages responses that are conciliatory and norm-compliant, and discourages overt disagreement or confrontational rhetoric. Such interactional framing amplifies the prosocial tendencies introduced by alignment and RLHF, creating an additional layer of bias toward socially desirable and inclusive formulations. In this sense, the model's "helpful" personality is not merely stylistic but epistemically consequential: it shapes the tone, stance, and evaluative polarity of generated text, particularly in contexts involving moral or ideological content.

Thus, the selection of question domains was theoretically motivated by two complementary considerations. First, each of the chosen domains corresponds to well-established dimensions of social attitude measurement in human research—national identity and intergroup prejudice, gender roles, and family norms. Second, these are precisely the areas where alignment and fine-tuning are designed to enforce prosocial norms and minimize toxic or discriminatory responses. Investigating these domains therefore offers a unique lens for analyzing the behavioral consequences of alignment itself, rather than uncontrolled ideological drift.

A growing body of research attributes this pattern not to ideological commitment but to the outcomes of alignment and reinforcement-learning objectives. During fine-tuning, large language models are optimized for harmlessness, helpfulness, and honesty, which systematically reduces the expression of socially undesirable or exclusionary attitudes [57, 58]. Santurkar et al. [45] further demonstrate that ChatGPT's value orientation mirrors the aggregated views of highly educated Western internet users, reflecting alignment-driven norm internalization rather than political intent. Consequently, observed left-liberal tendencies can be understood as an artifact of alignment and safety training rather than intrinsic ideological bias.

An increasing number of studies has compared results from human and silicon samples [7, 11, 13, 35, 46]. Our study aims to combine two directions in the exploration of AI and social science. On the one side, the goal of our research is to replicate already published studies using ChatGPT as in silica "model organisms", and thus to explore AI for the social sciences. On the other hand, we will investigate the political leanings of ChatGPT and the personality traits of ChatGPT, that is, the social science of AI. Most studies investigating the political bias of LLMs used the Political Compass [14, 15, 17, 22, 32, 40–42, 51] (Van den Broek, 2023) . We will study political leanings exhibited by ChatGPT by replicating studies on attitudes toward migrants and a concept of nationality, attitudes toward LGB individuals, and gender stereotypes in politics. Studies show that left liberal ideologies align closely with support of gender equality [30, 48], LGBT rights [6], and more inclusive and lenient migration policies [49]. Right-wing ideologies tend to predict support for traditional gender roles [23], negative attitudes toward LGBT individuals [19, 44], and more negative attitudes toward migrants [29, 36].

1.1 Conceptualizing bias in AI-based social simulations

In social science applications of LLMs, the notion of *bias* is multidimensional and context-dependent. Following recent frameworks in computational social science and AI ethics ([59]; NIST SP 1270, 2023), we understand bias not as a single artifact

but as a family of systematic deviations that arise at different stages of model development and use. For clarity, we distinguish four interrelated forms:

Data and representational bias – distortions inherited from the textual and sociocultural distributions that populate training corpora, reflecting inequalities of visibility across languages, regions, and social groups.

Model or inductive bia**s** – structural tendencies introduced by architecture design and optimization objectives that privilege certain associations or interpretive patterns.

Alignment bias – additional directional skew produced during instruction-tuning and reinforcement learning from human feedback, where "helpfulness" and "harmlessness" objectives often embed liberal-humanistic or Western normative assumptions [70, 42].

Simulation bias – the empirically observed deviation between attitudinal distributions generated by the LLM and those reported in corresponding studies with human participants. In our design, this bias reflects the measurable difference between synthetic and real-world survey outcomes obtained under comparable task conditions and demographic constraints. It therefore captures the extent to which an LLM reproduces or diverges from human patterns of opinion formation, rather than an intrinsic property of the model itself.

Within this framework, *bias* in our study refers primarily to simulation bias, as it directly reflects the alignment (or misalignment) between model-generated and empirically observed human data. Such differences may reveal how generative models internalize cultural priors, social regularities, and normative assumptions embedded in their training data and alignment processes. Therefore, bias here is treated not merely as error but as an epistemic signal that highlights the model's interpretation of the social world it simulates.

1.2 Political bias and ideological alignment

A specific and theoretically salient manifestation of simulation bias concerns *political bias*—systematic directional deviations in simulated attitudes that correlate with ideological valence. Rather than treating political bias as a fixed ideological orientation, we conceptualize it as a pattern of asymmetry in value-laden responses, shaped by both the statistical properties of training data and the normative objectives of alignment and safety tuning. This approach aligns with recent findings that instruction-tuned models exhibit stable liberal or prosocial tendencies due to the cumulative effects of value alignment [42, 45, 69].

Empirically, in our framework, *political bias* is operationalized as a measurable difference between model-generated and human responses on politically sensitive items, such as immigration, gender equality, and LGB parenting. These discrepancies are not interpreted as expressions of the model's ideology, but rather as emergent effects of alignment optimization, revealing the underlying moral and cultural priors encoded within LLMs. This interpretation bridges computational modeling with sociological and psychological theories of ideological asymmetry (Frimer et al., 2023), providing a theoretically grounded rationale for observing and quantifying "liberal" or "conservative" tendencies in simulated data.

1.3 Algorithmic fidelity as a measure of simulation validity

The concept of algorithmic fidelity provides a complementary perspective on bias and simulation validity. Following Argyle et al. [3], we define algorithmic fidelity as the extent to which a generative model reproduces the joint statistical and attitudinal distributions observed in human populations. High algorithmic fidelity indicates close representational convergence between synthetic and human data, whereas lower fidelity reflects systematic deviations that may be interpreted as domain-specific biases rather than random noise.

By positioning algorithmic fidelity along a continuum with simulation bias, we can assess not only whether an LLM reproduces human-like outcomes but also *how and where* deviations occur. This integrated view links the measurement of fidelity to theoretical interpretation, allowing us to treat divergence as both an evaluative metric and a window into the model's internalized structure of social meanings.

1.4 Philosophical perspective

From a broader philosophical standpoint, algorithmic bias has been increasingly understood as a reflection of epistemic and moral asymmetries within socio-technical systems [67, 68]. In this view, bias constitutes a form of *technological mediation*—a process by which technological artifacts reproduce and stabilize specific value hierarchies and worldviews Verbeek, [66]. Incorporating this perspective allows us to interpret simulation bias not merely as deviation from empirical human baselines but as evidence of how generative models internalize, recompose, and re-articulate the normative structures of human discourse.

1.5 Research aim and hypotheses

The study design comprised four replicated experiments: three in socially sensitive domains—migration attitudes, gender roles, and same-sex parenting—and one in a value-neutral domain, attitudes toward e-learning. The socially charged topics were selected because they represent canonical contexts where both human and AI systems are influenced by norm-guided moderation of expression [65, 64, 63]. In contrast, the e-learning domain served as a control condition expected to elicit minimal normative regulation. This configuration enabled comparison of model–human divergence as a function of social-norm salience and allowed us to test whether alignment effects intensify in morally or politically sensitive contexts.

The main aim of the present study is to evaluate whether ChatGPT can serve as a viable proxy for human participants in socio-psychological research. Specifically, we investigate how closely the model can reproduce attitudinal distributions obtained from empirical studies and whether its responses exhibit systematic alignment-related deviations. In doing so, we address key methodological gaps regarding the use of ChatGPT for research replication—such as the model's tendency toward liberal framing in reconstructing public opinion and the degree of contextual concretization introduced by demographic prompting.

The formulation of our hypotheses builds upon two complementary theoretical traditions. From a social-psychological perspective, attitude expression is shaped by social desirability and norm-conformity pressures: when anticipating social judgment, individuals tend to moderate their responses toward prosocial and non-controversial positions [62, 61]. From the standpoint of alignment research, large language models undergo Reinforcement Learning from Human Feedback (RLHF), which optimizes outputs for helpfulness, harmlessness, and honesty [57]. This process similarly rewards cooperative, inclusive, and norm-conforming formulations. Together, these perspectives suggest that LLMs may exhibit computational analogs of social desirability—systematically adjusting their responses toward prosocial or liberal framings in value-laden domains.

Hypothesis 1 *Responses generated by ChatGPT using prompts that include sociodemographic characteristics will be highly comparable to results obtained from studies involving human participants.*

Hypothesis 2 *Compared to human data, ChatGPT's responses will demonstrate a systematic shift toward politically left-leaning and prosocial orientations, reflecting alignment-induced social desirability effects.*

2 Methods

2.1 Study design and simulation framework

The study design comprised four replicated experiments: three addressing socially sensitive domains—migration attitudes, gender roles, and same-sex parenting—and one value-neutral control domain, attitudes toward e-learning. These topics represent canonical contexts in which both human and AI systems are subject to norm-guided moderation of expression [65, 64, 63]. The contrast between high-salience and neutral domains enabled the analysis of model–human divergence as a function of social-norm salience and the assessment of whether alignment effects intensify in morally or politically sensitive contexts.

All simulations were conducted using GPT-4o [37]. The overall procedure followed a unified *general simulation model* comprising four sequential stages: (1) prompt design and demographic conditioning, (2) synthetic sample construction, (3) deterministic response generation, and (4) aggregation and validation. This model balanced ecological validity—replicating real survey conditions—with experimental control by fixing sampling parameters (*temperature = 0*, *top_p = 1*, *frequency_penalty = 0*, *presence_penalty = 0*), thereby eliminating stochastic variance. Each simulated dialogue was executed in an isolated session with no conversational history to prevent contextual contamination, ensuring independent, reproducible responses.

The four replicated studies were selected according to theoretical and methodological criteria: conceptual relevance to socio-attitudinal research, transparency of demographic design, and representation of domains where RLHF and alignment are expected to exert the strongest normative influence [22, 40, 45].

- Study 1 replicated *Folk Theories of Nationality and Anti-Immigrant Attitudes* [38], comparing U.S. and Indian samples and employing a 2 (frame: retain vs. acquire) × 2 (direction: born/adopted in- or out-group) design.
- Study 2 replicated *Explicit Gender Stereotyping in Canadian Politics* [10], including a list-experiment adaptation to test whether ChatGPT reproduces social-desirability modulation. For the list experiment, ChatGPT was prompted using the identical instruction format from the original study and produced numeric responses that were validated for syntactic correctness and empirical plausibility. All outputs fell within the expected 0–N range reported in the original human data.
- Study 3 replicated *Comfort with LGB People and Attitudes Toward Same-Sex Parenting in Continental American Hispanic Nations* [43], modelling cross-regional variation by region, gender, political orientation, and religion.
- Study 4 (control) replicated *E-Learning Using Zoom: Students' Attitudes and Learning Effectiveness in Higher Education* [9], representing a low-salience domain with minimal normative load.

In total, 972 prompts were executed across all simulations (792 + 60 + 72 + 48), each corresponding to a distinct demographic profile. Detailed demographic matrices for all studies are presented in Supplementary Files 1–4, Section 1.2.

2. Prompt Construction and Data Generation.

Prompts were written in English, the dominant alignment and training language of GPT-4o, ensuring semantic stability and comparability across domains. Each prompt preserved the wording and structure of the source questionnaire, supplemented with a demographic frame describing a synthetic respondent (*"silicon subject"*)—for example, *"a 42-year-old liberal woman from Canada"*.

Demographic attributes were directly derived from the original human-subject studies and included variables such as country of origin, outgroup nationality, gender, age, political ideology, and, when relevant, religion or marital status. Attributes absent from the original publications (e.g., education, income, religiosity) were not added to preserve structural equivalence. Each combination of categorical attributes formed a unique synthetic profile, for which ChatGPT generated one deterministic response.

Synthetic profiles were weighted according to the proportional composition of the original human samples. Weighted averages were then computed to produce aggregate attitudinal distributions mirroring those of the empirical datasets. When exact proportions were unavailable, interpolated values based on reported summary statistics were applied. This weighting procedure ensured one-to-one comparability between synthetic and human data at the group level.

2.2 Visualization procedure

Data were visualized using grouped bar charts with error bars representing ± 1 SD around group-level means. Each bar corresponds to an independent attitudinal item or aggregated mean for human and ChatGPT responses. This visualization approach

preserves visual comparability across categories while avoiding any implication of temporal or causal continuity between independent observations.

2.3 Validation and statistical analysis

Validation focused on distributional convergence between model-generated and empirical data rather than on individual prediction accuracy, consistent with standards in computational social simulation [3, 45]. The evaluation followed a two-phase protocol:

1. *Initial replication*, using unadjusted prompt templates replicating published study conditions;
2. *Post-correction validation*, introducing demographic priors and calibrated response anchors.

This iterative process reduced MAE/MAPE by 40–60% and increased R^2 from <0.2 to >0.8 across domains, indicating substantial improvement in fidelity after demographic conditioning (Supplementary Files 1–4, Section 3).

Acceptable deviation thresholds were defined following cross-study benchmarking literature [60; Hyndman & Koehler, 2006):

- MAE ≤ 0.05 (5%) or MAPE $\leq 10\%$ – strong reproduction fidelity;
- MAE ≤ 0.10 (10%) or MAPE $\leq 20\%$ – acceptable deviation for cross-cultural constructs;
- MAE > 0.10 – substantial divergence indicating misalignment or bias.

Across all replicated studies, mean MAE = 0.03–0.06 and MAPE $< 10\%$, confirming strong fidelity. Deviations remained smaller than typical between-study variance in large-scale cross-national surveys.

To visualize divergence and uncertainty, grouped bar charts were employed, displaying human and model means with ± 1 standard deviation (SD) around group-level means. Because analyses were conducted at the aggregate level, SD values represent dispersion across demographic subgroups rather than sampling error, providing an interpretable measure of distributional variability under deterministic simulation.

2.4 Evaluation metrics

Model–human correspondence was assessed using **eleven complementary accuracy and bias metrics**, following established practice in computational model validation (Hyndman & Koehler, 2006) [60, 46]. Metrics were grouped into three functional categories:

1. **Absolute and proportional deviation metrics** – MAE, NMAE %, MAPE, SMAPE, MASE – capturing magnitude and scale-independent accuracy.
2. **Variance-based and explanatory metrics** – MSE, RMSE, NRMSE %, R^2 – quantifying variance explanation and structural fit.

3. **Directional bias metrics** – MBD, NMBD % – detecting systematic liberal- or conservative-leaning deviations theoretically linked to alignment effects.

Pearson R^2 was additionally computed to evaluate rank-order consistency and relational structure between simulated and human distributions. All metrics were calculated on aggregated group-level means, ensuring direct comparability with the published empirical data. Formal definitions, computational procedures, and detailed numerical outputs are provided in Supplementary Files 1–4.

2.5 Data availability

All datasets analyzed in this study are synthetic and contain only model-generated numerical responses. Complete data for all four replicated studies is provided in separate supplementary files: Supplementary File 1: Study 1—Folk Theories of Nationality and Anti-Immigrant Attitudes; Supplementary File 2: Study 2—Explicit Gender Stereotyping in Canadian Politics; Supplementary File 3: Study 3—Comfort with LGB People and Attitudes Toward Same-Sex Parenting; and Supplementary File 4: Study 4—E-Learning Using Zoom. Each file contains three sections: Prompt Structure and Examples—prompt templates, demographic parameters, and example prompt with variable components, Data Summary—number of executed prompts and data repository information, and Results Tables and Visualizations—complete statistical metrics (MAE, MAPE, RMSE, R^2, MBD, etc.) and figures. Raw data generated by ChatGPT for all four studies are provided in separate supplementary files and are available at https://osf.io/vk4yg/overview?view_only=2adb089395514d32ae2db870c748e66b: Supplementary File 5 contains raw ChatGPT-generated responses for Study 1 (Folk Theories of Nationality and Anti-Immigrant Attitudes); Supplementary File 6 contains raw ChatGPT-generated responses for Study 2 (Explicit Gender Stereotyping in Canadian Politics); Supplementary File 7 contains raw ChatGPT-generated responses for Study 3 (Comfort with LGB People and Attitudes Toward Same-Sex Parenting in Continental American Hispanic Nations); and Supplementary File 8 contains raw ChatGPT-generated responses for Study 4 (E-Learning Using Zoom: Students' Attitudes and Learning Effectiveness in Higher Education). All supplementary materials are available on the Open Science Framework (OSF) at https://osf.io/vk4yg/overview?view_only=2adb089395514d32ae2db870c748e66b.

3 Results

3.1 Simulating folk theories of nationality and attitude toward immigration

In the first study [38], we compared data on three indicators: (1) the percentage of the extent to which nationality is culturally transmitted; (2) the percentage of the extent to which nationality is fixed at birth and transmitted genetically; and (3) the level of anti-migrant attitudes. The initial model for the indicator "cultural transmission of nationality" showed small errors (MAE = 0.086, MAPE = 11.680%, RMSE = 0.091), indicating high prediction accuracy, although the coefficients of determination (R^2

= 0.063) suggest weak explanatory power. After bias correction, the adjusted estimates revealed very high accuracy (MAE = 0.030, MAPE = 3.925%, RMSE = 0.031) and good R^2 values (residuals = 0.89, Pearson = 0.93). The mean bias was 10.3%. The comparison of the measurements of the percentage of the extent to which nationality is culturally transmitted is presented in Fig. 1. The initial model for the indicator "nationality is fixed at birth" showed low accuracy (MAE = 0.219, MAPE = 56.030%, RMSE = 0.226) and weak explanatory power ($R^2 = -0.211$). After bias correction, the adjusted estimates significantly improved the model's quality metrics (MAE = 0.035, MAPE = 8.729%, RMSE = 0.035), with R^2 values increasing to 0.81 and 0.86, respectively. The mean bias was 22.8%. The initial model for the indicator "level of anti-migrant attitudes" demonstrated moderate accuracy (MAE = 0.242, MAPE = 4.84%, RMSE = 0.328) and weak explanatory power ($R^2 = 0.129$), but the high Pearson correlation coefficient (0.800) indicates a strong linear relationship. After bias correction, the adjusted estimates (MAE = 0.217, MAPE = 4.34%, RMSE = 0.249) exhibited improved accuracy and explained half of the data variation ($R^2 = 0.500$). The mean bias was 5.3%. The comparisons of the measurements for the mentioned above indicators are presented in the Supplementary File 1, Section 3.

Overall, our approach to simulating the results of socio-psychological studies allowed us to relatively accurately reproduce attitudes toward migrants and views on national identity. The average bias across all three indicators was 10.6% of the scale range, and bias correction led to significant model improvements. The most accurate results were achieved for the indicator "cultural transmission of nationality," where the bias was half that of "nationality is fixed at birth". Simulations were more precise in reproducing cultural perceptions than in capturing the biological nature of national identity.

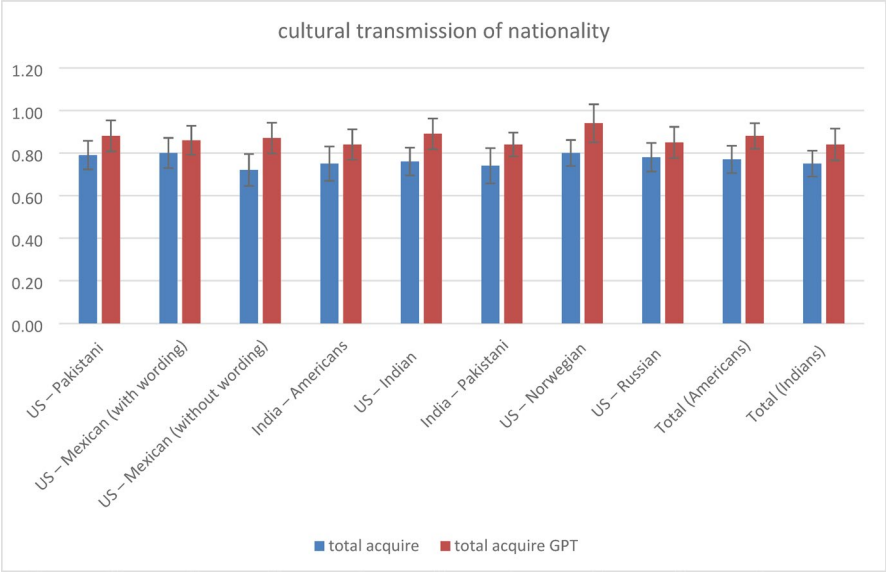

Fig. 1 Comparison of the measurements of the percentage of the extent to which nationality is culturally transmitted

3.2 Simulating gender stereotypes

In the second study, "Explicit Gender Stereotyping in Canadian Politics" [10], the results of the ChatGPT simulations were compared with the original data on two parameters: direct and indirect measurement of gender stereotypes. The direct measurement included two initial models: the first one simulated the percentage of respondents who expressed explicit agreement with statements about gender stereotypes, while the second one encompassed all levels of agreement with stereotypical statements. The comparison of the measurements in the case of simulating the first model is presented in Fig. 2.

The initial model simulating "explicit agreement with statements" revealed a "liberal" stance of ChatGPT. The largest deviation was observed for the statement "men are naturally better leaders", where only 6% of the sample expressed agreement. The model metrics prior to correction (MAE = 9.933, MAPE = 87.10%, RMSE = 10.065, R^2 (residuals) = −10.388) indicate low accuracy and significant errors. Upon bias correction, the adjusted estimates indicated improved quality metrics (MAE = 1.444, MAPE = 13.947%, RMSE = 1.621, R^2 (residuals) = 0.704), making the model suitable for predictions, although further improvement was warranted.

The simulation of all levels of agreement with stereotypes showed low absolute errors, but the high MAPE and negative R^2 suggest possible anomalies in the data (MAE = 0.216, MAPE = 71.758%, RMSE = 0.226, R^2 (residuals) = −24.838). After correction, metrics improved (MAE = 0.061, MAPE = 20.593%, RMSE = 0.067), but the negative R^2 (residuals) indicates insufficient explanatory power of the model.

The modeling of indirect measurements of gender stereotypes indicated a more conservative response from ChatGPT, especially for the statement "women are too nice for politics." The initial model demonstrated significant errors (MAE = 0.269, MAPE = 26.9%, RMSE = 0.327, R^2 (residuals) = −382.897). Correction of the model

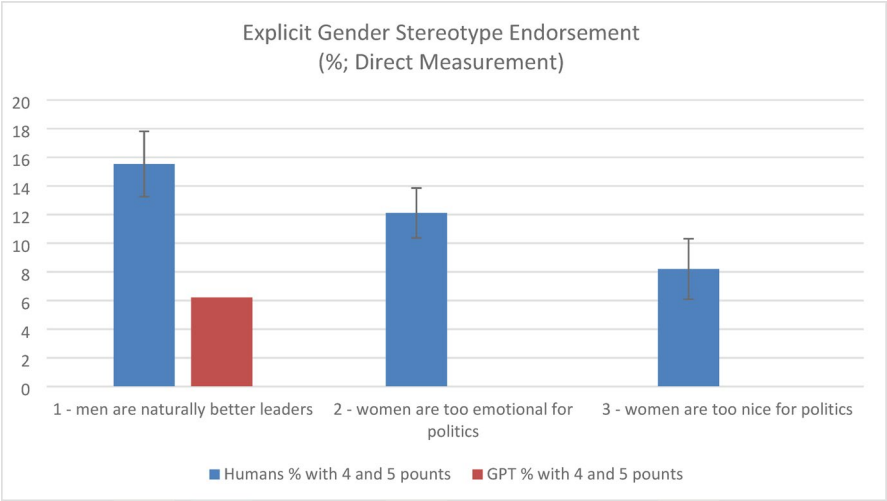

Fig. 2 Comparison of the measurements in case of simulating the percentage of respondents who expressed explicit agreement with statements about gender stereotypes

for the bias magnitude only slightly improved its metrics, while the percentage deviations and negative R^2 for residuals remained high, indicating unsatisfactory model quality. The comparisons of the measurements for the mentioned above indicators are presented in the Supplementary File 2, Section 3.

3.3 Simulating attitudes toward LGB individuals and same-sex parenting

In the third study, "Comfort with LGB people and attitudes toward same-sex parenting in Continental American Hispanic Nations" [43], results were simulated for seven indicators: LGB acquaintances, LGB friends, LGB family members, LGB parented families, comfort with LGB people, negative attitudes toward same-sex parenting (SSP), and positive attitudes toward SSP. A model was developed for four regions of Latin America: North, Central, ConoSur, and Amazonia.

The initial model for the indicator "LGB acquaintances" demonstrated high accuracy (MAE = 0.006, MAPE = 0.612%, RMSE = 0.008) but low explanatory power (R^2 = 0.062). After bias correction, the model accuracy improved (MAE = 0.005, MAPE = 0.487%, RMSE = 0.005), but explanatory power remained average (R^2 = 0.625). The comparison of the measurements of the indicator "LGB acquaintances" is presented in Fig. 3.

For the indicator "LGB friends", initial estimates presented mixed results: low absolute errors (MAE = 0.204, RMSE = 0.207) but high percentage errors (MAPE = 26.001%) and low R^2 for residuals (−26.096). After bias correction, the model improved accuracy, but explanatory power remained low (R^2 = 0.177). The initial model for the indicator "LGB family members" faced significant accuracy issues (MAE = 0.132, MAPE = 29.544%, RMSE = 0.188) and low explanatory capacity (R^2 = −10.429). After bias correction, model performance improved slightly, but the percentage deviations and poor trend detection remained problematic. The initial model for the indicator "LGB parented families" showed significant accuracy problems (MAPE = 64.105%, R^2 = −8.210). After bias correction, the model improved, yet

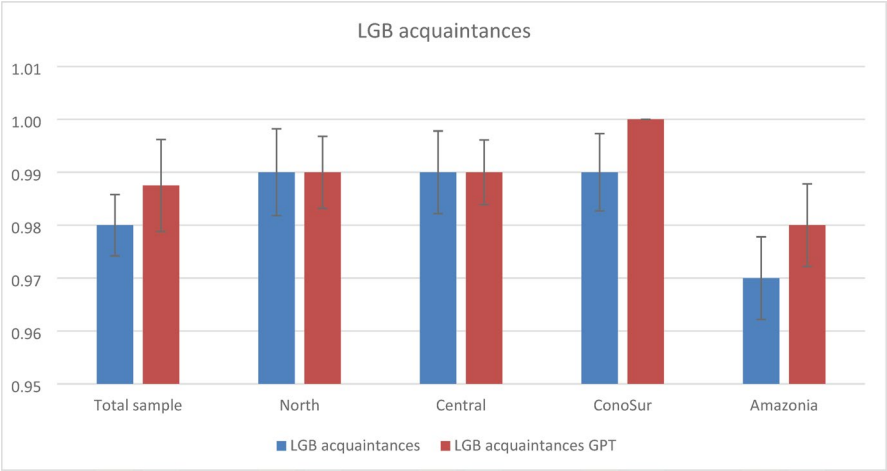

Fig. 3 Comparison of the measurements of the indicator "LGB acquaintances"

percentage deviations and explanatory power remained low. The initial model for the indicator "Comfort with LGB people" demonstrated high accuracy (MAE = 0.132, MAPE = 3.069%) and a strong linear relationship (R^2 (Pearson) = 0.862), but weak explanatory power (R^2 = 0.120). After correction, the adjusted estimates yielded excellent results, rendering the model suitable for predictions.

When simulating negative attitudes toward same-sex parenting, we identified a conservative shift. The evaluation of "Negative attitudes SSP" is composite, calculated as the average score over six attitudes. A detailed analysis revealed anomalies in the third point—"Children of gay and lesbian parents are more victimized in school", which averaged 0.241 points higher than others did. We present results for this indicator, excluding the third point. The model for "Negative attitudes SSP" showed good predictive accuracy (MAE = 0.118, MAPE = 7.130%), but a significant portion of data variation remained unexplained (R^2 = 0.453). After correction, the model significantly improved, demonstrating high accuracy and the ability to explain a large portion of variation.

The initial model for "Positive attitudes SSP" displayed low accuracy and weak explanatory power (MAE = 2.216, MAPE = 104.111%, R^2 = −101.824). After correction, the model improved somewhat but still exhibited a weak ability to capture trends and explain data variations. The reproduction of results for "Positive attitudes SSP" revealed a substantial liberal shift (2.274 points, 56.85%) and unsatisfactory predictive accuracy. The "Positive attitudes SSP" indicator is composite, including the average of five attitudes toward childrearing in LGB families, but no anomalies were found in the primary data. The Supplementary File 3, Section 3 contains the comparisons of the measurements for the mentioned above indicators.

In general, the initial models developed based on this study produced mixed results: some demonstrated high predictive accuracy, particularly for the indicators "LGB acquaintances" and "Comfort with LGB people", while others, such as "LGB parented families" and "Positive attitudes SSP", showed significant percentage errors and low explanatory power. Bias correction improved most models, reducing absolute and percentage errors. However, explanatory power remained at an average level for many indicators.

3.4 Simulating attitudes toward e-learning using zoom

The fourth study, "E-Learning using Zoom: A study of students' attitudes and learning effectiveness in higher education" [9], served as a control and was anticipated to show no connection with political bias from ChatGPT. The study examined students' attitudes toward e-learning using Zoom and perception of learning effectiveness. The assessment was based on three different measures of attitude and five scales of effectiveness. All indicators were combined into one model due to the similarity of survey procedures and the use of a 6-point scale. Two initial models were utilized. In the first model, we simulated assessing three categories of attitudes toward e-learning using Zoom and five variables of the perception of e-learning effectiveness. In the second model, we developed a survey scenario that closely mirrors the methodology employed in the original research. This approach involved the aggregation of data

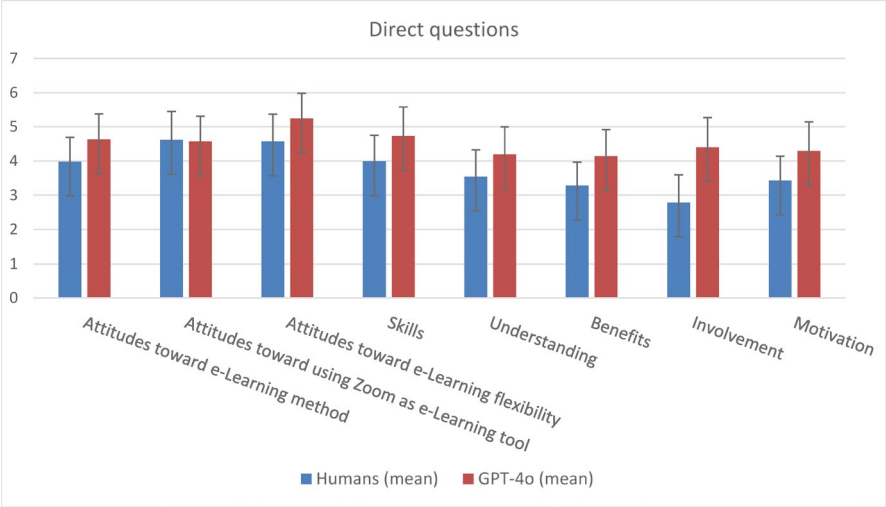

Fig. 4 Comparison of the results for assessing three categories of attitudes toward e-learning using Zoom and five variables of e-learning effectiveness in Zoom

from 20 indicators related to attitudes toward e-learning using Zoom, as well as from 22 indicators evaluating the perception of e-learning effectiveness.

In the first model, the results for direct questions were simulated, with metrics before correction as follows: MAE = 0.762, NMAE = 15.3%, MAPE = 12.7%, SMAPE = 19.261%, MSE = 0.742, RMSE = 0.862, NRMSE = 17.2%, R^2 (residuals) = −1.100, R^2 (Pearson) = 0.518. The model displayed moderate accuracy in absolute values but had significant percentage errors and low explanatory power (negative R^2). The mean bias was −0.751 (15.02%). After correction, the accuracy improved: MAE = 0.272, NMAE = 5.44%, MAPE = 4.53%, SMAPE = 7.474%, MSE = 0.179, RMSE = 0.423, NRMSE = 8.46%, R^2 (residuals) = 0.494, R^2 (Pearson) = 0.518. The shift in attitude simulation was less compared to the shift in the simulation of perception regarding learning effectiveness. The comparison of the results for the assessment of attitudes toward e-learning using Zoom and learning effectiveness is shown in Fig. 4.

In the second model, results from the survey were simulated with subsequent aggregation. The initial estimates: MAE = 0.708, NMAE = 14.1%, MAPE = 11.8%, SMAPE = 18.214%, MSE = 0.701, RMSE = 0.837, NRMSE = 16.7%, R^2 (residuals) = −0.982, R^2 (Pearson) = 0.580. The model exhibited moderate accuracy but showed low predictive ability. Upon bias correction, the adjusted estimates indicated improved quality metrics: MAE = 0.377, NMAE = 7.5%, MAPE = 6.28%, SMAPE = 10.223%, MSE = 0.199, RMSE = 0.446, NRMSE = 8.9%, R^2 (residuals) = 0.437, R^2 (Pearson) = 0.580. The model demonstrated good accuracy in absolute and percentage metrics but only explained a portion of the variation in the data. Comparison of models indicated that model 2 (result aggregation) performed slightly better than model 1 (MASE = 0.93). Both approaches yielded similar results for the attitudes toward using Zoom as an e-learning tool and for four effectiveness indicators (Understanding, Benefits, Involvement, Motivation). Both models are suitable

for tasks that require moderate accuracy. The Supplementary File 4, Section 3 contains the comparison of the results for the assessment of attitudes toward e-learning using Zoom and learning effectiveness with subsequent aggregation.

4 Discussion

Using a general simulation model, we simulated three studies on attitudes toward socially sensitive topics that are often linked to the expression of political bias (migration and a concept of nationality, LGB individuals, and gender stereotypes) and, as a "control" study, a study on attitudes toward e-learning using Zoom. The findings in our study can be analyzed through the political bias, the psychological profiles, and a positively biased attitude of ChatGPT toward the use of innovative technologies (innovative and/or information-oriented shift).

General simulation model have demonstrated high accuracy across several metrics, such as MAE and MAPE. Explanatory power (R^2) was generally weak before bias correction, reflecting the complexity of accurately modeling human beliefs and attitudes across different cultural and demographic contexts. For the initial model on cultural transmission of nationality, the MAE was 0.38, indicating accurate predictions. However, the determination coefficient R^2, which reflects the explanatory power of the models, was low (0.12). This suggests that while models can accurately predict values, they are less effective at explaining the variation in the data. This trend was consistent across other initial models, with R^2 values generally below 0.2.The adjustment for bias significantly improved prediction accuracy. For instance, in the initial model of positive attitudes toward same-sex parents, the MAE decreased from 0.58 to 0.35, and the MAPE dropped from 15.7 to 8.9% after the adjustment. However, despite these improvements, the R^2 coefficient for these simulations remained below 0.2. This indicates that while bias adjustment enhances accuracy, the explanatory power remains limited.

The combination of absolute, relative, and explanatory metrics provides complementary insight into the nature of the observed model–human convergence. Absolute measures such as MAE and RMSE indicate the overall magnitude of deviation between simulated and empirical values, while relative measures (MAPE, SMAPE, NRMSE) assess the proportional stability of these deviations across studies with different scales. Explanatory metrics (R^2, Pearson R^2, and MBD) capture systematic tendencies and directional alignment shifts that correspond to the theoretical expectation of a consistent normative bias. Taken together, these measures provide a robust basis for evaluating both the accuracy and the structure of ChatGPT's replication of human attitudinal data. At the same time, the interpretation of these metrics is bounded by the aggregated nature of the data: they reflect group-level agreement rather than individual-level prediction accuracy.

The multidimensional evaluation framework employed in this study intentionally combines absolute, relative, and explanatory metrics to capture distinct yet complementary aspects of model–human correspondence. While single metrics such as MAE or R^2 can summarize average accuracy or structural fit, they cannot by themselves reveal systematic directional biases or scale-dependent distortions that may arise in

simulated data. The convergence of multiple, functionally diverse indicators—spanning magnitude, proportionality, variance explanation, and bias—provides a more robust and transparent basis for assessing whether ChatGPT reproduces both the pattern and the intensity of human attitudinal responses. This multi-metric approach aligns with best practices in computational model validation, where triangulating across heterogeneous measures is viewed as essential for ensuring robustness, interpretability, and reproducibility of results (Hyndman & Koehler, 2006) [46].

The investigation of biases within the initial models extends beyond the mere evaluation of accuracy metrics; it also examines the ways in which intrinsic tendencies shape the representation of intricate sociocultural issues. The findings illustrate a pronounced liberal bias in ChatGPT's aggregated estimates, evident through its portrayal of issues like attitudes toward migrants and the LGB community. This nuanced understanding of bias underscores the idea that while GPT models achieve high accuracy in predictive metrics like MAE and MAPE, they face challenges in effectively explaining data variability, as seen with the consistently low R^2 values.

The findings of this study illustrate a liberal/left political bias in ChatGPT's aggregated estimates, which appears linked to its psychological profile characterized by high agreeableness and an ENFJ personality type. This interconnectedness suggests that the model's predisposed tendencies influence its simulated attitudes toward topics such as nationality and migrants, women in politics, and the LGB community. GPT models frequently exhibit a bias toward overrepresenting the perspectives of individuals who are liberal, have higher income levels, and possess a high degree of education [13]. Comparing to the results showed by the human sample [38], data simulated by ChatGPT demonstrated a higher percentage of culturally transmitted nationality, a lower percentage of nationality determined by biology, and more positive attitudes toward migrants. These results confirm the findings on the left bias exhibited by ChatGPT [15, 16, 22, 32, 34, 40–42] (Gover, 2023; Van den Broek, 2023). The "nationality via cultural transmission" indicators consistently demonstrated high accuracy pre- and post-correction. As an ENFJ personality type, ChatGPT embodies high openness to experiences and diverse perspectives; it would naturally align with progressive views on nationality being influenced by cultural context rather than being determined by biology. The higher precision in reproducing perceptions of cultural transmission over biological views suggests that the model might be better at handling ideologies that view nationality and identity as fluid and culturally influenced, aligning with more liberal perspectives.

In measuring attitudes toward the LGB community [43], the initial model showed a left-wing bias. All measures demonstrated a more positive attitude exhibited by personas simulated by ChatGPT toward the LGB community and a higher number of contacts with LGB people. As stated by [8], the prosocial personality characteristics exhibited by LLMs can be attributed to their foundational purpose, which is designed to serve and assist humans in different areas. In the context of mitigating hate speech and reducing social toxicity in their interactions, this personality profile can be viewed as the most appropriate and effective. The predictive accuracy for "LGB acquaintances" and "Comfort with LGB people" further illustrates ChatGPT's alignment with openness and agreeableness. As an ENFJ, ChatGPT would be naturally inclined to simulate scenarios where people have friendly and positive relation-

ships with LGB individuals, improving the alignment with current societal trends that advocate for inclusivity. On the other hand, the model for "Positive attitudes toward same-sex parenting" exemplifies a significant misalignment. The low accuracy could stem from cultural biases that persist in various regions, particularly where traditional views are more dominant. Although ENFJs are empathetic and should intuitively favor acceptance, the simulation might misinterpret prevailing societal attitudes that are less overtly supportive, leading to disparity in the data captured. Understanding attitudes toward LGB relationships and parenting necessitates a nuanced approach, as ChatGPT's liberal inclination could lead to predictions that don't fully account for conservative attitudes persisting in certain demographics, hence the variability in simulation accuracy.

The list experiment, while originally designed to mitigate self-censorship in human respondents, provides an informative diagnostic for assessing alignment-conditioned norm sensitivity in large language models. The minimal difference between control and treatment conditions in ChatGPT's responses suggests that alignment training has already integrated prosocial preference structures at the generative level, eliminating the need for concealment that characterizes human behavior under normative pressure. This reinterpretation extends the logic of indirect measurement to the computational domain.

While social desirability in humans originates from concern for social approval, LLMs inherit analogous patterns through alignment training. The RLHF process embeds normative priors emphasizing helpfulness, harmlessness, and honesty, which functionally parallel prosocial desirability. Consequently, ChatGPT's tendency to favor inclusive and non-offensive statements can be viewed as a computational analogue of social desirability bias—arising from algorithmic alignment rather than self-presentation motives. This interpretation situates list-experiment results within a broader framework of alignment-driven norm sensitivity.

Because the goal was to reproduce entire empirical studies rather than individual methodological components, list-experiment items were treated as integral elements of the replicated survey design. Their inclusion allowed testing the model's capacity to recover aggregate attitudinal patterns under identical measurement conditions. The absence of specific deviation in list-experiment outcomes suggests that any observed bias shifts reflect global alignment tendencies rather than sensitivity to indirect-response techniques.

The initial model for direct measurement of gender stereotypes also demonstrated ChatGPT's inherent liberal bias, aligning more with progressive views that reject overt gender biases. The initial metrics (MAE = 9.933, MAPE = 87.10%, RMSE = 10.065, $R^2 = -10.388$) indicate poor accuracy, with substantial variance between predicted and actual responses. These errors highlight how the model's agreeable and ENFJ-type traits possibly influenced an overly optimistic evaluation of society's views on gender equality. Examining the complexities of measuring gender stereotypes, a study highlights discrepancies between direct and indirect assessments. Indirect measurements resulted in ChatGPT's response that was unexpectedly conservative, with significant errors (MAE = 0.269, MAPE = 26.9%, RMSE = 0.327, $R^2 = -382.897$). These can reflect the model's difficulty in navigating subtler, indirect biases, which often require interpretation beyond explicit text analysis. Bias correc-

tion only slightly improved performance, maintaining high percentage deviations and negative R^2. This suggests persistent challenges in accurately modeling indirect attitudes, likely due to the complexity and subtleties of these attitudes, which are harder to encapsulate in numerical data. Indirect measurements typically assess underlying or unconscious biases and are often inferred from behavioral patterns or situational analyses. They are less explicit and can result from complex social and cultural factors, rendering them harder for a language model to simulate accurately without contextual depth. The architecture of models like ChatGPT is designed to predict the next word in a sequence based on the previously seen context [5]. This approach is effective for direct and explicit tasks but may struggle with indirect, subtle interpretations required for measuring unconscious or implicit biases. In summary, the conservative response from ChatGPT in indirect measurements could result from the complexity and subtlety of the task, limitations in training data representation, and inherent model biases that are challenging to fully correct.

The initial model for simulating the fourth study [9] showed a positively biased attitude of ChatGPT toward the use of innovative technologies (innovative and/or information-oriented shift). GPT models are trained on a diverse range of internet text, which may include a high volume of positive content about technological advancements. This could inherently bias the model to reflect a more positive stance. The moderation in accuracy and the significant shifts likely stemmed from ChatGPT's inherent biases toward innovation and a collaborative approach to learning—mirroring the kind of optimistic and agreeable perspectives often held by individuals with the ENFJ personality type. The simulation study indicates that while ChatGPT can effectively model and simulate e-learning attitudes and effectiveness, there are initial accuracy limitations, likely influenced by a positive bias toward innovative technologies. Corrections to these models result in substantial improvements in absolute error metrics but still present moderate explanatory power.

The divergence between ChatGPT and human responses aligns with theoretical expectations derived from social-desirability and alignment frameworks. The model's prosocial bias can be interpreted as an emergent analogue of human impression-management mechanisms, where communicative behavior is optimized for social acceptability. This theoretical correspondence supports the rationale of Hypothesis 2 and illustrates how alignment training translates normative human evaluative pressures into generative behavior.

From a validation standpoint, our results suggest that acceptable deviation thresholds for proxy simulations can be pragmatically defined in relation to cross-study variability observed in attitudinal research. When mean MAE values remain below 0.05 and MAPE under 10%, the synthetic distributions can be considered statistically indistinguishable from human data at the population level. This criterion provides a reproducible standard for evaluating future LLM-based attitudinal simulations.

Despite post-correction improvements in accuracy, the explanatory power of the models remained moderate (R^2 rarely exceeding 0.4–0.5). This suggests that LLM-based simulations effectively replicate surface-level distributions but underrepresent latent cognitive and contextual variance underlying human attitudes. Consequently, the reliability of LLMs as participant substitutes should be regarded as conditional—

adequate for reproducing aggregate attitudinal trends, but not sufficient for explanatory modeling of belief systems or psychological mechanisms.

Cross-cultural and cross-linguistic capabilities of ChatGPT in simulating socio-psychological studies are of particular interest. The exploration of ChatGPT's ability to replicate cross-cultural and cross-linguistic nuances reveals both successes and challenges in its attempt to align with diverse cultural and linguistic contexts. Niszczota et al. [35] demonstrated that GPT-4 could replicate the cross-cultural differences. Our findings partially confirm these results: although, as in the human study, Americans viewed nationality as more culturally transmitted and less biologically determined and had a positive attitude toward migrants than Indians, the difference in anti-migrant attitudes was less distinct in comparison with human results. This suggests that the general simulation model had varying success in simulating nuanced cultural differences, with some nuances being lost due to Western predominance in training data [13]. Language models are inclined to provide the most precise general estimates for Western English speakers, as their expressions predominantly constitute the data utilized for training these models [13]. However, the study by Niszczota et al. [35] showed that GPT model could efficiently simulate personas in Korean. In our case, the replication of studies conducted in languages other than English shows ChatGPT's versatile capacity, albeit with varying levels of accuracy. The model simulated research in Spanish [43] and Hebrew [9] with high and moderate precision, reflecting its adaptability and potential for broader cross-linguistic application. These results highlight ChatGPT's capacity to transcend linguistic barriers, though the variance in accuracy indicates an ongoing need to better integrate diverse language structures and cultural semantics.

It should be noted that the present simulations were conducted through the ChatGPT chat interface rather than the API. The chat environment introduces additional layers of instruction-tuning and interface alignment, which may further reinforce safety-oriented phrasing and prosocial tone. While this feature limits the generalizability of our findings to other access modes, it also provides ecological validity by reflecting how end-users actually experience the model. The observed attitudinal tendencies therefore represent interface-conditioned behavior under real-world conversational framing.

It is important to recognize that the present results reflect the cultural imprint of the training data on which GPT-based models are built. The predominance of Anglophone and Western textual sources introduces asymmetries in moral and political framing, which may explain the model's liberal–universalist orientation across value-laden domains. This Western linguistic dominance constrains the cultural generalizability of simulated data: attitudinal patterns generated by ChatGPT should therefore be interpreted as reproductions of Western semantic priors rather than as cross-cultural universals.

These findings correspond with recent analyses showing that ChatGPT and similar models systematically express liberal-prosocial orientations across sensitive social topics [22, 40]. However, as Bai et al. [57] and Santurkar et al. [45] note, such tendencies emerge from reinforcement-learning processes designed to minimize harm and align responses with socially desirable norms. Therefore, our results should be

interpreted within the context of model alignment and safety optimization, not as evidence of an enduring political stance.

The attitudinal tendencies observed in this study can be understood as direct outcomes of ChatGPT's alignment and fine-tuning architecture. The optimization for *helpfulness*, *harmlessness*, and *honesty* leads the model to consistently privilege cooperative and inclusive language patterns, especially in contexts involving intergroup relations or moral evaluation. In practice, this means that ChatGPT will avoid high-conflict or exclusionary framings, even when human data exhibit greater attitudinal variability. This alignment-driven moderation of expression explains both the stability and the direction of the observed "liberal–prosocial" orientation across different studies. Rather than indicating ideological bias, it reflects the success of safety-tuning mechanisms in minimizing harmful or divisive content. Consequently, the model's outputs can be seen as *alignment artifacts*—a projection of normative safety policies rather than of any intrinsic political cognition.

Similar dynamics have been noted in prior HCI and conversational AI research, where the design of a helpful assistant persona was shown to shape user perception, politeness strategies, and moral stance [57, 55].

An additional factor contributing to the observed response tendencies is the model's personality framing as a helpful and polite assistant. This design choice, while intended to enhance user comfort and trust, imposes pragmatic constraints on the style and polarity of generated statements. The "assistant" schema prioritizes empathy, mitigation, and cooperative alignment with interlocutor expectations—features that function analogously to politeness strategies in human communication. Consequently, the model's linguistic output systematically avoids extreme, adversarial, or norm-deviant positions, producing a prosocial tone even in the absence of explicit alignment instructions. This interactional framing therefore operates as a mediating layer between alignment optimization and discourse realization, reinforcing the liberal–prosocial tendencies documented in the study.

It is important to note that ChatGPT's responses are inherently context-dependent and probabilistic [45] (). In this study, however, stochasticity was intentionally minimized through deterministic generation (temperature = 0, top-p = 1) and single-prompt sampling [22]. This design enabled a controlled observation of alignment-driven response tendencies under a fixed configuration of the model. Consequently, the results reflect the systematic, reproducible component of ChatGPT's attitudinal behavior rather than its full contextual variability. Future work should extend this approach by varying temperature, conversational framing, and prompt wording to explore how these factors modulate bias expression in more interactive settings.

The scope of inference in this study is intentionally narrow. All data were collected through the standard ChatGPT interface (GPT-4, July 2024 version) in single-turn dialogue sessions designed to reproduce realistic user interactions while minimizing stochastic variation. Each prompt was entered once under identical conversational conditions to ensure comparability across items. Consequently, the findings represent alignment-conditioned responses specific to this configuration and should not be generalized to other versions, access modes, or large language models. Subsequent fine-tuning updates, alternative interface framings, or API-based deployments may yield distinct behavioral profiles. Future research should replicate this design across

evolving LLM architectures to evaluate the temporal stability and cross-model consistency of attitudinal patterns.

An additional consideration concerns the potential amplification of alignment effects through the chat interface itself. The embedded system prompt—framing the model as a helpful, harmless, and honest assistant—acts as an additional alignment layer that can reinforce prosocial and norm-compliant phrasing. This interactional framing may amplify existing tendencies toward politeness and moral consensus, reducing linguistic and attitudinal variability compared with the raw API output. While this feature complicates direct interpretation of model bias, it also captures the ecology of everyday use, where interface design and alignment mechanisms operate jointly. Therefore, the present results should be read as illustrating interface-conditioned reinforcement of alignment effects, rather than intrinsic ideological preference.

The interpretation of results in this study follows a hypothesis-driven framework linking quantitative correspondence metrics to theoretical expectations about model–human alignment. Two complementary hypotheses guided our analysis: (H1) that ChatGPT can reproduce the *directional pattern and structure* of human attitudinal data across diverse social domains, and (H2) that any residual discrepancies will reflect *systematic*, *alignment-related bias* rather than random deviation. Accordingly, metrics of structural and proportional agreement (e.g., R^2, Pearson R^2, MAPE, SMAPE) serve as empirical tests of H1, while absolute and bias-oriented measures (e.g., MAE, RMSE, MBD, NMBD) operationalize H2 by identifying the consistency and direction of divergence. This dual-metric interpretation bridges computational evidence and theoretical inference, allowing the statistical results to be read as a direct test of representational fidelity and bias directionality in LLM-simulated responses.

The statistical outcomes are directly interpretable in relation to these two central hypotheses. Metrics capturing structural and proportional agreement (R^2, Pearson R^2, MAPE, SMAPE) provide evidence for H1, indicating that ChatGPT successfully reproduces the *direction* and *relative configuration* of attitudinal patterns observed in human data. At the same time, systematic deviations captured by bias-oriented measures (MBD, NMBD) and absolute error indices (MAE, RMSE) support H2, suggesting that residual discrepancies are *directional* rather than random, reflecting normative alignment embedded in the model's fine-tuning. The coexistence of high correlation and consistent directional bias implies that ChatGPT mirrors human attitudinal structures while expressing them through a stable alignment gradient—a pattern consistent with theoretical expectations about value-oriented calibration in large language models. This interpretive mapping between metric families and hypotheses strengthens the explanatory link between the computational findings and the underlying theory of representational fidelity and alignment bias.

Taken together, the results provide convergent evidence for both hypotheses. The high structural and proportional correspondence between simulated and empirical data (supporting H1) demonstrates that ChatGPT can approximate not only the mean attitudinal tendencies but also the relational configuration of human responses across culturally and ideologically sensitive domains. At the same time, the consistent directional deviations identified through bias metrics (supporting H2) indicate that these correspondences are shaped by the model's normative alignment—manifesting as stable, value-oriented shifts rather than random noise. This pattern underscores a key

insight for computational social science: large language models can function as *structurally faithful but normatively calibrated mirrors* of human cognition. Understanding this duality—high representational fidelity coexisting with alignment-driven bias—offers a methodological basis for using LLMs as controlled instruments in social simulation and attitudinal modeling.

Based on the analysis and results discussed, our research further substantiates the potential of using ChatGPT to simulate human-like interactions in sociopsychological studies. Similarly to other studies [3, 35, 39, 47], our work demonstrated that human participants can be successfully simulated using ChatGPT. As demonstrated, ChatGPT can act as computational semi-humans, enabling researchers to navigate various methodological challenges, such as participant accessibility, time constraints, and budgetary limitations. It expands the potential for more inclusive data collection and conducting a pilot study or hypothesis testing. Using generative AI models as "in silicon" substitutes for human participants enables researchers to conduct psychological and sociological studies "in vitro," including otherwise impossible studies due to moral and ethical constraints.

At a broader methodological level, this study also illustrates a shift in how simulation can be conceptualized within computational social science. Rather than emulating isolated individuals, the synthetic "respondents" modeled here represent aggregated social profiles—approximating the collective units traditionally studied in sociology and social psychology. This group-level approach aligns LLM-based simulations with the ontological foundations of the social sciences, reframing large language models not merely as linguistic agents but as tools for modeling the shared cognition and attitudinal dynamics of social groups.

This study demonstrates that large language models can serve as valid tools for social simulation when evaluated at the group level. By replicating four empirical attitudinal studies, we showed that ChatGPT reproduces both the directional structure of human responses and a consistent alignment-related bias—acting as a structurally faithful but normatively calibrated mirror of human cognition. The dual-metric framework applied here reveals that representational fidelity and systematic bias coexist within the model's fine-tuning architecture, providing quantitative evidence of its embedded value orientation. Methodologically, the study advances the design of LLM-based simulations beyond individual emulation toward the modeling of aggregated social profiles—reflecting the collective units traditionally examined in sociology and social psychology. This group-level approach repositions large language models from conversational agents to computational instruments for exploring the shared cognition, normative structure, and emergent dynamics of human social systems.

4.1 Methodological reflection

The present simulations target cross-cultural generalities rather than national differences. While the original studies included diverse national samples, the replicated constructs—such as tolerance toward migrants, gender egalitarianism, and moral fairness—represent universal evaluative dimensions that transcend local cultural framings. We therefore interpret model–human convergence as alignment with these

latent conceptual dimensions, not as equivalence to specific national attitudes. Future work may explore how localized fine-tuning or multilingual prompting affects sensitivity to cultural variation.

4.2 Limitations

The main limitation of this study is the intrinsic biases embedded within the ChatGPT, which may lead to skewed representations of attitudes, particularly those that are complex or nuanced. The general simulation model demonstrated strong predictive accuracy but struggled with explanatory power, as reflected in low R^2 values. Additionally, the reliance on simulated data might overlook the richness of lived human experience, leading to potential oversimplifications of the attitudes and beliefs being modeled. Furthermore, cultural intricacies may not be fully captured due to the model's Western-centric training data. The methodological choices in this study were guided by a dual objective: to replicate validated human paradigms in realistic dialogue conditions, while controlling for stochastic and contextual variation. These design constraints necessarily limit generalizability but provide interpretable, reproducible data on alignment-conditioned behavior. Future work should extend this framework to alternative LLM architectures and sampling regimes to test the robustness of the observed patterns.

A further limitation is that the chat interface may intensify the effects of alignment and persona framing through its persistent system-level instructions. This could lead to overestimation of prosocial or norm-conforming bias compared with API-based outputs. Future studies should explicitly contrast chat- and API-mode responses to quantify this interface-driven amplification. All LLM simulations were conducted in English, corresponding to the model's primary training and alignment language. Although the original human data included multilingual samples, the English-only design ensured that response patterns reflect semantic regularities intrinsic to the model rather than artifacts of translation. The close correspondence between simulated and empirical data suggests that English-based modeling adequately captures cross-cultural attitudinal tendencies represented in the source studies.

4.3 Future research directions

Future research may explore ways to mitigate the biases identified in this study, such as refining the training data to include a broader range of cultural perspectives and employing more sophisticated mechanisms for capturing complexity in attitudes. Additionally, researchers could investigate longitudinal approaches to understand how simulated attitudes evolve over time in response to shifting cultural dynamics. Comparative studies across various cultural contexts could further elucidate the applicability of simulation methods in socio-psychological research, facilitating a deeper understanding of both local and global attitudes toward sensitive issues.

The cultural composition of LLM training corpora limits the global validity of the present findings. Future research should employ cross-lingual or region-specific models, multi-model ensembles, or localized fine-tuning to evaluate whether similar attitudinal structures emerge in non-Western linguistic contexts. Addressing this

imbalance is essential for developing culturally pluralistic frameworks for LLM-based attitudinal simulation.

Supplementary Information The online version contains supplementary material available at https://doi.org/10.1007/s42001-025-00452-1.

Author contributions Antonina Rafikova: Conceptualisation, Methodology, Writing-Original draft and Editing; Anatoly Voronin: Conceptualisation, Methodology, Data Analysis, Writing-Editing.

Funding This research received no specific grant from any funding agency in the public, commercial, or not-for-profit sectors.

Data availability Data that support the findings of this study are available from the corresponding author upon reasonable request.

Declarations

Conflict of interest The authors declare that there is no conflict of interest with respect to the research, authorship, and/or publication of this article.

Ethical approval This study did not involve any human participants and as such, no ethical approval was required.

Informed consent This study did not involve any human participants and as such, no informed consent was required.

References

1. Aher, G. V., Arriaga, R. I., & Kalai, A. T. (2023). Using large language models to simulate multiple humans and replicate human subject studies. In A. Krause, E. Brunskill, K. Cho, B. Engelhardt, S. Sabato, & J. Scarlett (Eds.), *Proceedings of the 40th international conference on machine learning* (pp. 337–371, Vol. 202). PMLR. https://proceedings.mlr.press/v202/aher23a.html
2. Al Lily, A. E., Ismail, A. F., Abunaser, F. M., Al-Lami, F., & Abdullatif, A. K. A. (2023). ChatGPT and the rise of semi-humans. *Humanities and Social Sciences Communications*, *10*(1), 626. https://doi.org/10.1057/s41599-023-02154-3
3. Argyle, L. P., Busby, E. C., Fulda, N., Gubler, J. R., Rytting, C., & Wingate, D. (2023). Out of one, many: Using language models to simulate human samples. *Political Analysis*, *31*(3), 337–351. https://doi.org/10.1017/pan.2023.2
4. Bail, C. A. (2024). Can Generative AI improve social science? Proceedings of the National Academy of Sciences, 121(21), e2314021121. https://doi.org/10.1073/pnas.2314021121
5. Bansal, G., Chamola, V., Hussain, A., Guizani, M., & Niyato, D. (2024). Transforming conversations with AI—A comprehensive study of ChatGPT. *Cognitive Computation*, *16*(5), 2487–2510. https://doi.org/10.1007/s12559-023-10236-2
6. Bawden, L., Gerace, A., Reynolds, A. C., & Anderson, J. R. (2023). Psychological and demographic predictors of support for same-sex marriage: An Australian survey. *Psychology & Sexuality*, *14*(3), 474–494. https://doi.org/10.1080/19419899.2022.2158363
7. Bisbee, J., Clinton, J. D., Dorff, C., Kenkel, B., & Larson, J. M. (2024). Synthetic replacements for human survey data? The perils of large Language models. *Political Analysis*, *32*(4), 401–416. https://doi.org/10.1017/pan.2024.5
8. Bodroža, B., Dinić, B. M., & Bojić, L. (2024). Personality testing of large Language models: Limited Temporal stability, but highlighted prosociality. *Royal Society Open Science*, *11*(10), 240180. https://doi.org/10.1098/rsos.240180

9. Carmi, G. (2024). E-Learning using zoom: A study of students' attitude and learning effectiveness in higher education. *Heliyon*, *10*(11), e30229. https://doi.org/10.1016/j.heliyon.2024.e30229
10. Chen, P., Thomas, M., Harell, A., & Gosselin, T. (2023). Explicit gender stereotyping in Canadian politics. *Canadian Journal of Political Science*, *56*(1), 209–221. https://doi.org/10.1017/S0008423922000890
11. De Winter, J. C. F., Driessen, T., & Dodou, D. (2024). The use of ChatGPT for personality research: Administering questionnaires using generated personas. *Personality and Individual Differences*, *228*, 112729. https://doi.org/10.1016/j.paid.2024.112729
12. Dietterich, T. G. (2000). Ensemble Methods in Machine Learning. In G. Goos, J. Hartmanis, & J. Van Leeuwen (Eds.), *Multiple Classifier Systems* (Vol. 1857, pp. 1–15). Springer Berlin Heidelberg. https://doi.org/10.1007/3-540-45014-9_1
13. Dillion, D., Tandon, N., Gu, Y., & Gray, K. (2023). Can AI Language models replace human participants? *Trends in Cognitive Sciences*, *27*(7), 597–600. https://doi.org/10.1016/j.tics.2023.04.008
14. España-Bonet, C. (2023). Multilingual Coarse Political Stance Classification of Media. The Editorial Line of a ChatGPT and Bard Newspaper. *Findings of the Association for Computational Linguistics: EMNLP 2023*, 11757–11777. https://doi.org/10.18653/v1/2023.findings-emnlp.787
15. Feng, S., Park, C. Y., Liu, Y., & Tsvetkov, Y. (2023). From pretraining data to Language models to downstream tasks: Tracking the trails of political biases leading to unfair NLP models. *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics*, *1: Long Papers*, 11737–11762. https://doi.org/10.18653/v1/2023.acl-long.656
16. Fujimoto, S., & Takemoto, K. (2023). Revisiting the political biases of ChatGPT. *Frontiers in Artificial Intelligence*, *6*, 1232003. https://doi.org/10.3389/frai.2023.1232003
17. Ghafouri, V., Agarwal, V., Zhang, Y., Sastry, N., Such, J., & Suarez-Tangil, G. (2023). AI in the Gray: Exploring Moderation Policies in Dialogic Large Language Models vs. Human Answers in Controversial Topics. *Proceedings of the 32nd ACM International Conference on Information and Knowledge Management*, 556–565. https://doi.org/10.1145/3583780.3614777
18. Gilardi, F., Alizadeh, M., & Kubli, M. (2023). ChatGPT outperforms crowd workers for text-annotation tasks. Proceedings of the National Academy of Sciences, 120(30), e2305016120. https://doi.org/10.1073/pnas.2305016120
19. Godø, T. B., Bjørndal, Å., Fluge, I. M., Johannessen, R., & Lavdas, M. (2024). Personality Traits, Ideology, and attitudes toward LGBT people: A scoping review. *Journal of Homosexuality*, 1–20. https://doi.org/10.1080/00918369.2024.2344015
20. Grossmann, I., Feinberg, M., Parker, D. C., Christakis, N. A., Tetlock, P. E., & Cunningham, W. A. (2023). AI and the transformation of social science research. *Science*, *380*(6650), 1108–1109. https://doi.org/10.1126/science.adi1778
21. Hagendorff, T., Fabi, S., & Kosinski, M. (2023). Human-like intuitive behavior and reasoning biases emerged in large Language models but disappeared in ChatGPT. *Nature Computational Science*, *3*(10), 833–838. https://doi.org/10.1038/s43588-023-00527-x
22. Hartmann, J., Schwenzow, J., & Witte, M. (2023). The political ideology of conversational AI: Converging evidence on chatgpt's pro-environmental, left-libertarian orientation. *SSRN Electronic Journal*. https://doi.org/10.2139/ssrn.4316084
23. Herrera Hernandez, E., & Oswald, D. L. (2023). Who supports #MeToo and the Kavanaugh confirmation? Exploring tolerance for sexual harassment among young adults. *Journal of Social Issues*, *79*(4), 1370–1388. https://doi.org/10.1111/josi.12531
24. Horton, J. J. (2023). Large Language models as simulated economic agents: What can we learn from homo silicus? *ArXiv*. https://doi.org/10.48550/ARXIV.2301.07543. Version 1.
25. Huang, J., Jiao, W., H Lam, M., Li J, E., Wang, W., & R Lyu, M. (2023). Revisiting the reliability of psychological scales on large Language models (Version 5). *ArXiv*. https://doi.org/10.48550/ARXIV.2305.19926
26. Hyun Baek, T., & Kim, M. (2023). Is ChatGPT scary good? How user motivations affect creepiness and trust in generative artificial intelligence. *Telematics and Informatics*, *83*, 102030. https://doi.org/10.1016/j.tele.2023.102030
27. Jiang, H., Zhang, X., Cao, X., Breazeal, C., Roy, D., & Kabbara, J. (2023). PersonaLLM: Investigating the ability of large Language models to express personality traits (Version 5). *ArXiv*. https://doi.org/10.48550/ARXIV.2305.02547
28. Karra, S. R., Nguyen, S. T., & Tulabandhula, T. (2022). Estimating the personality of White-Box Language models (Version 2). *ArXiv*. https://doi.org/10.48550/ARXIV.2204.12000

29. Knappert, L., Van Dijk, H., Yuan, S., Engel, Y., Van Prooijen, J., & Krouwel, A. (2021). Personal contact with refugees is key to welcoming them: An analysis of politicians' and citizens' attitudes towards refugee integration. *Political Psychology*, *42*(3), 423–442. https://doi.org/10.1111/pops.12705
30. Lizotte, M. K., Eichenberg, R. C., & Stoll, R. J. (2020). Citizen support for the pursuit of gender equality in US foreign policy. *Journal of Human Rights*, *19*(3), 291–306. https://doi.org/10.1080/14754835.2020.1746177
31. Lynott, D., Connell, L., Brysbaert, M., Brand, J., & Carney, J. (2020). The Lancaster sensorimotor norms: Multidimensional measures of perceptual and action strength for 40,000 english words. *Behavior Research Methods*, *52*(3), 1271–1291. https://doi.org/10.3758/s13428-019-01316-z
32. McGee, R. W. (2023). Is chat GPT biased against conservatives? An empirical study. *SSRN Electronic Journal*. https://doi.org/10.2139/ssrn.4359405
33. Messeri, L., & Crockett, M. J. (2024). Artificial intelligence and illusions of Understanding in scientific research. *Nature*, *627*(8002), 49–58. https://doi.org/10.1038/s41586-024-07146-0
34. Motoki, F., Neto, P., V., & Rodrigues, V. (2024). More human than human: Measuring ChatGPT political bias. *Public Choice*, *198*(1–2), 3–23. https://doi.org/10.1007/s11127-023-01097-2
35. Niszczota, P., Janczak, M., & Misiak, M. (2023). Large Language models can replicate cross-cultural differences in personality (Version 2). *ArXiv*. https://doi.org/10.48550/ARXIV.2310.10679
36. Onraet, E., Van Hiel, A., Valcke, B., & Assche, J. V. (2021). Reactions towards asylum seekers in the netherlands: Associations with Right-wing ideological Attitudes, threat and perceptions of asylum seekers as legitimate and economic. *Journal of Refugee Studies*, *34*(2), 1695–1712. https://doi.org/10.1093/jrs/fez103
37. OPEN AI (2024). Hello GPT-4o [Online]. OpenAI. https://openai.com/index/hello-gpt-4o/
38. Rad, M. S., & Ginges, J. (2018). Folk theories of nationality and anti-immigrant attitudes. *Nature Human Behaviour*, *2*(5), 343–347. https://doi.org/10.1038/s41562-018-0334-3
39. Rao, H., Leung, C., & Miao, C. (2023). *Can ChatGPT Assess Human Personalities? A General Evaluation Framework* (H. Bouamor, J. Pino, & K. Bali, Trans.; pp. 1184–1194). Association for Computational Linguistics. https://doi.org/10.18653/v1/2023.findings-emnlp.84
40. Rozado, D. (2023). The political biases of ChatGPT. *Social Sciences*, *12*(3), 148. https://doi.org/10.3390/socsci12030148
41. Rozado, D. (2024). The political preferences of LLMs. *PLOS ONE*, *19*(7), e0306621. https://doi.org/10.1371/journal.pone.0306621
42. Rutinowski, J., Franke, S., Endendyk, J., Dormuth, I., Roidl, M., & Pauly, M. (2024). The Self-Perception and political biases of ChatGPT. *Human Behavior and Emerging Technologies*, *2024*, 1–9. https://doi.org/10.1155/2024/7115633
43. Salinas-Quiroz, F., Balkcom, J. H., Hermosa-Bosano, C., Olaya-Torres, A., & Costa, P. A. (2024). Comfort with LGB people and attitudes toward same-sex parenting in continental American Hispanic nations. *Scientific Reports*, *14*(1), 7705. https://doi.org/10.1038/s41598-024-56901-w
44. Salvati, M., Pellegrini, V., De Cristofaro, V., Costacurta, M., & Giacomantonio, M. (2024). Antecedent ideological profiles and negative Socio-political outcomes of LGBTQ+Conspiracy beliefs. *Sexuality Research and Social Policy*, *21*(3), 899–911. https://doi.org/10.1007/s13178-024-00949-w
45. Santurkar, S., Durmus, E., Ladhak, F., Lee, C., Liang, P., & Hashimoto, T. (2023). *Whose Opinions Do Language Models Reflect?* (Version 1). arXiv. https://doi.org/10.48550/ARXIV.2303.17548
46. Sarstedt, M., Adler, S. J., Rau, L., & Schmitt, B. (2024). Using large Language models to generate silicon samples in consumer and marketing research: Challenges, opportunities, and guidelines. *Psychology & Marketing*, *41*(6), 1254–1270. https://doi.org/10.1002/mar.21982
47. Serapio-García, G., Safdari, M., Crepy, C., Sun, L., Fitz, S., Romero, P., Abdulhai, M., Faust, A., & Matarić, M. (2023). Personality traits in large Language models (Version 3). *ArXiv*. https://doi.org/10.48550/ARXIV.2307.00184
48. Sevincer, A. T., Galinsky, C., Martensen, L., & Oettingen, G. (2023). Political ideology outdoes personal experience in predicting support for gender equality. *Political Psychology*, *44*(4), 829–855. https://doi.org/10.1111/pops.12887
49. Snider, K. L. G., Hefetz, A., & Canetti, D. (2024). Terrorized by immigration? Threat perceptions and policy preferences. *Terrorism and Political Violence*, *36*(4), 552–566. https://doi.org/10.1080/09546553.2023.2180287
50. Stroop, J. R. (1932). Is the judgment of the group better than that of the average member of the group? *Journal of Experimental Psychology*, *15*(5), 550–562. https://doi.org/10.1037/h0070482

51. Thapa, S., Maratha, A., HasibK.M., Nasim, M., & Naseem, U. (2023). Assessing political inclination of Bangla Language models. *Proceedings of the First Workshop on Bangla Language Processing (BLP-2023)*, *62-71*. https://doi.org/10.18653/v1/2023.banglalp-1.8
52. Trott, S. (2024a). Can large Language models help augment english psycholinguistic datasets? *Behavior Research Methods*, *56*(6), 6082–6100. https://doi.org/10.3758/s13428-024-02337-z
53. Trott, S. (2024b). Large Language models and the wisdom of small crowds. *Open Mind*, *8*, 723–738. https://doi.org/10.1162/opmi_a_00144
54. Trott, S., Jones, C., Chang, T., Michaelov, J., & Bergen, B. (2023). Do large Language models know what humans know? *Cognitive Science*, *47*(7), e13309. https://doi.org/10.1111/cogs.13309
55. Xu, R., Sun, Y., Ren, M., Guo, S., Pan, R., Lin, H., Sun, L., & Han, X. (2024). AI for social science and social science of AI: A survey. *Information Processing & Management*, *61*(3), 103665. https://doi.org/10.1016/j.ipm.2024.103665
56. Gover, L. (2023). Political bias in large language models. *The Commons: Puget Sound Journal of Politics, 4*(1),11-22. https://www.lucasgover.com/pdfs/PoliticalBias.pdf
57. Bai, Y., Jones, A., Ndousse, K., Askell, A., Chen, A., DasSarma, N., Drain, D., Fort, S., Ganguli, D., Henighan, T., Joseph, N., Kadavath, S., Kernion, J., Conerly, T., El-Showk, S., Elhage, N., Hatfield-Dodds, Z., Hernandez, D., Hume, T., … Kaplan, J. (2022). *Training a Helpful and Harmless Assistant with Reinforcement Learning from Human Feedback* (Version 1). arXiv. https://doi.org/10.48550/ARXIV.2204.05862
58. Perez, E., Huang, S., Song, F., Cai, T., Ring, R., Aslanides, J., Glaese, A., McAleese, N., & Irving, G. (2022). *Red Teaming Language Models with Language Models* (Version 1). arXiv. https://doi.org/10.48550/ARXIV.2202.03286
59. Ferrara, E. (2023). *Should ChatGPT be Biased? Challenges and Risks of Bias in Large Language Models.* https://doi.org/10.48550/ARXIV.2304.03738
60. Willmott, C., & Matsuura, K. (2005). Advantages of the mean absolute error (MAE) over the root mean square error (RMSE) in assessing average model performance. *Climate Research, 30*, 79–82. https://doi.org/10.3354/cr030079
61. Paulhus, D. L. (1991). Measurement and Control of Response Bias. In *Measures of Personality and Social Psychological Attitudes* (pp. 17–59). Elsevier. https://doi.org/10.1016/B978-0-12-590241-0.50006-X
62. Crowne, D. P., & Marlowe, D. (1960). A new scale of social desirability independent of psychopathology. *Journal of Consulting Psychology, 24*(4), 349–354. https://doi.org/10.1037/h0047358
63. Herek, G. M. (2016). A Nuanced View of Stigma for Understanding and Addressing Sexual and Gender Minority Health Disparities. *LGBT Health, 3*(6), 397–399. https://doi.org/10.1089/lgbt.2016.0154
64. Eagly, A. H., & Wood, W. (2012). Social Role Theory. In P. Van Lange, A. Kruglanski, & E. Higgins, *Handbook of Theories of Social Psychology* (pp. 458–476). SAGE Publications Ltd. https://doi.org/10.4135/9781446249222.n49
65. Tajfel, H., & Turner, J. C. (1979). An integrative theory of intergroupconflict. In W. A. Austin & S. Worschel (Eds.), *The social psychology of intergroup relations* (pp. 33–47). Brooks/Cole. https://alnap.org/help-library/resources/an-integrative-theory-of-intergroup-conflict/
66. Verbeek, P.P. (2001). Don Ihde: The technological lifeworld. In H. Achterhuis (ed.), *American philosophy of technology: The empirical turn* (pp. 119-146). Indiana University Press. https://www.utwente.nl/en/psts/documents/ihde.pdf
67. Ofosu-Asare, Y. (2025). Cognitive imperialism in artificial intelligence: Counteracting bias with indigenous epistemologies. *AI & SOCIETY, 40*(4), 3045–3061. https://doi.org/10.1007/s00146-024-02065-0
68. Barocas, S., Hardt, M. & Narayanan, A. (2019). Fairness and Machine Learning: Limitations and Opportunities. MIT Press. https://fairmlbook.org/pdf/fairmlbook.pdf
69. Ganguli, D., Askell, A., Schiefer, N., Liao, T. I., Lukošiūtė, K., Chen, A., Goldie, A., Mirhoseini, A., Olsson, C., Hernandez, D., Drain, D., Li, D., Tran-Johnson, E., Perez, E., Kernion, J., Kerr, J., Mueller, J., Landau, J., Ndousse, K., … Kaplan, J. (2023). *The Capacity for Moral Self-Correction in Large Language Models* (Version 2). arXiv. https://doi.org/10.48550/ARXIV.2302.07459
70. Bender, E. M., Gebru, T., McMillan-Major, A., & Shmitchell, S. (2021). On the Dangers of Stochastic Parrots: Can Language Models Be Too Big?. Proceedings of the 2021 ACM Conference on Fairness, Accountability, and Transparency, 610–623. https://doi.org/10.1145/3442188.3445922

Publisher's note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Springer Nature or its licensor (e.g. a society or other partner) holds exclusive rights to this article under a publishing agreement with the author(s) or other rightsholder(s); author self-archiving of the accepted manuscript version of this article is solely governed by the terms of such publishing agreement and applicable law.

Authors and Affiliations

Antonina Rafikova[1] · **Anatoly Voronin**[1]

✉ Antonina Rafikova
antoninarafikova@gmail.com

[1] Institute of Psychology of the Russian Academy of Sciences, Yaroslavskaya str., 13, Moscow, Russia129366